Enterprise Database Infrastructure Administration Lab

Design, Deployment, Security Hardening, Backup, Replication and Monitoring of PostgreSQL Infrastructure on Linux

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SUMMARY

LINUX INFRASTRUCTURE PREPARATION

I. Objective

The purpose of this deliverable is to prepare the infrastructure that will host the PostgreSQL environment.

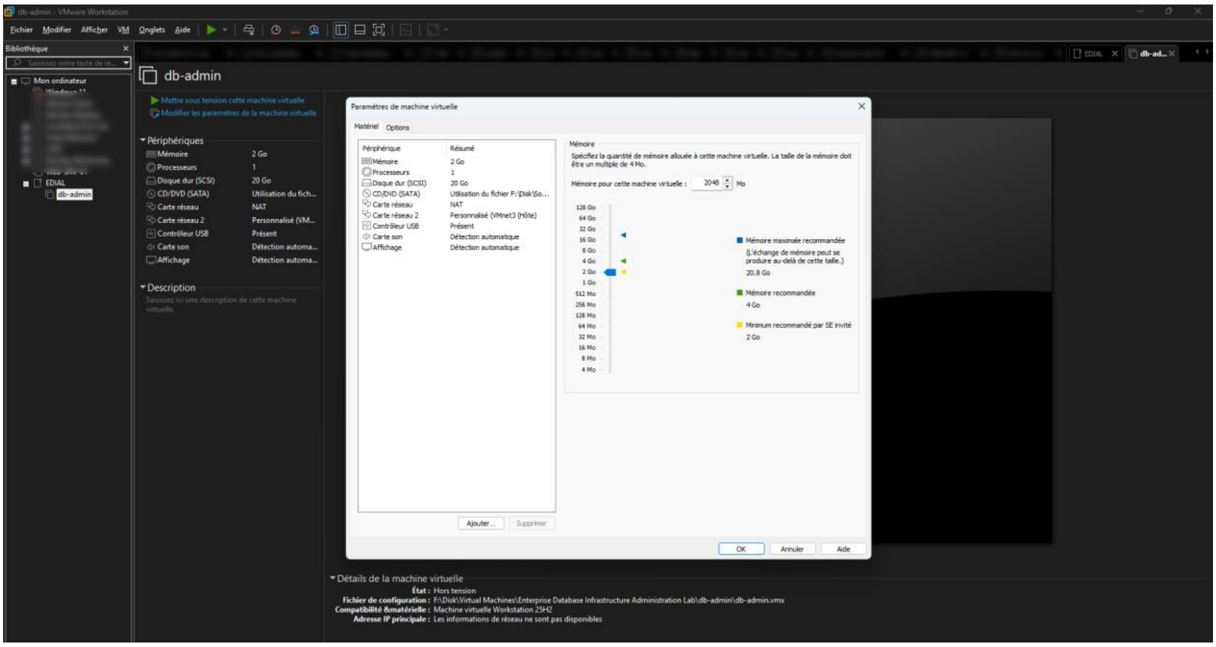
At the end of this phase :

- All virtual machines will be deployed
- Static IP addressing will be configured
- Hostnames will be configured
- SSH administration will be operational
- Administrative accounts will be created
- Basic system hardening will be completed
- All servers will communicate properly

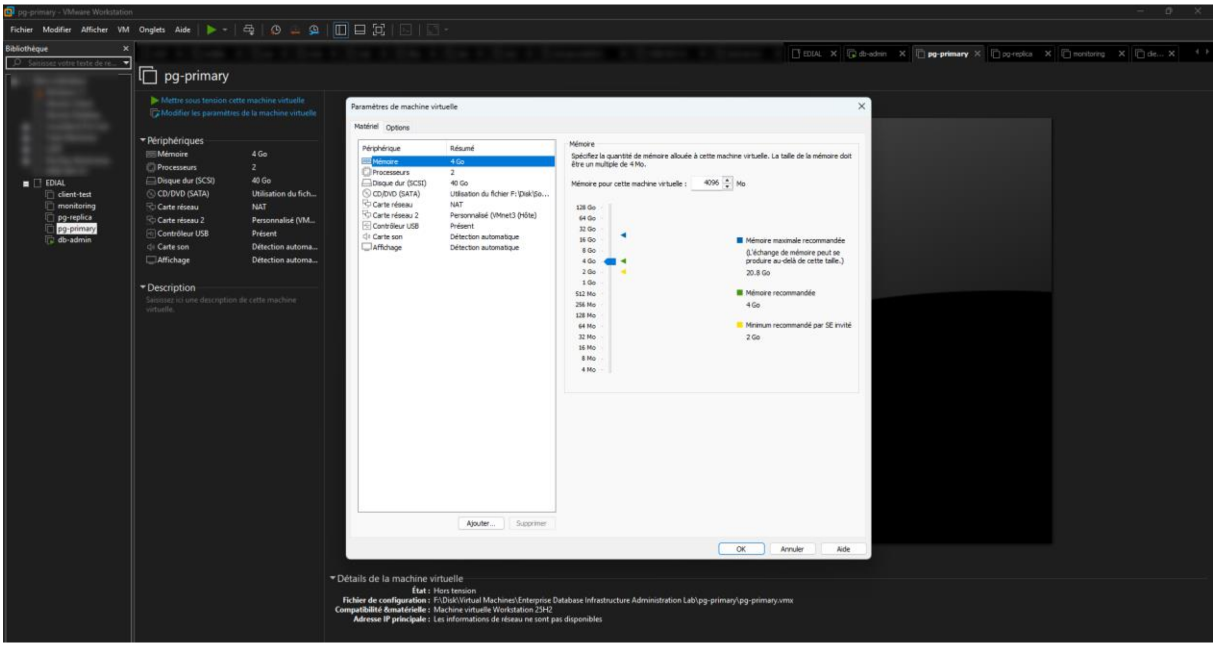
This deliverable establishes the foundation upon which the entire database infrastructure will be built.

II. Infrastructure Deployment

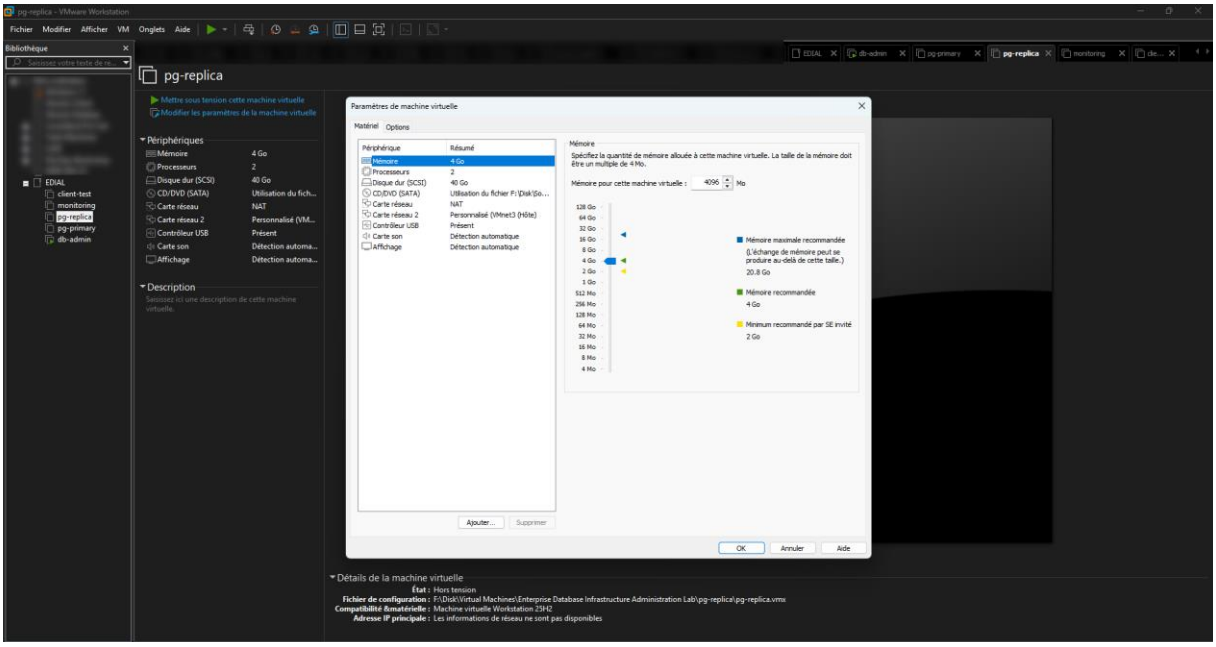
1. db-admin



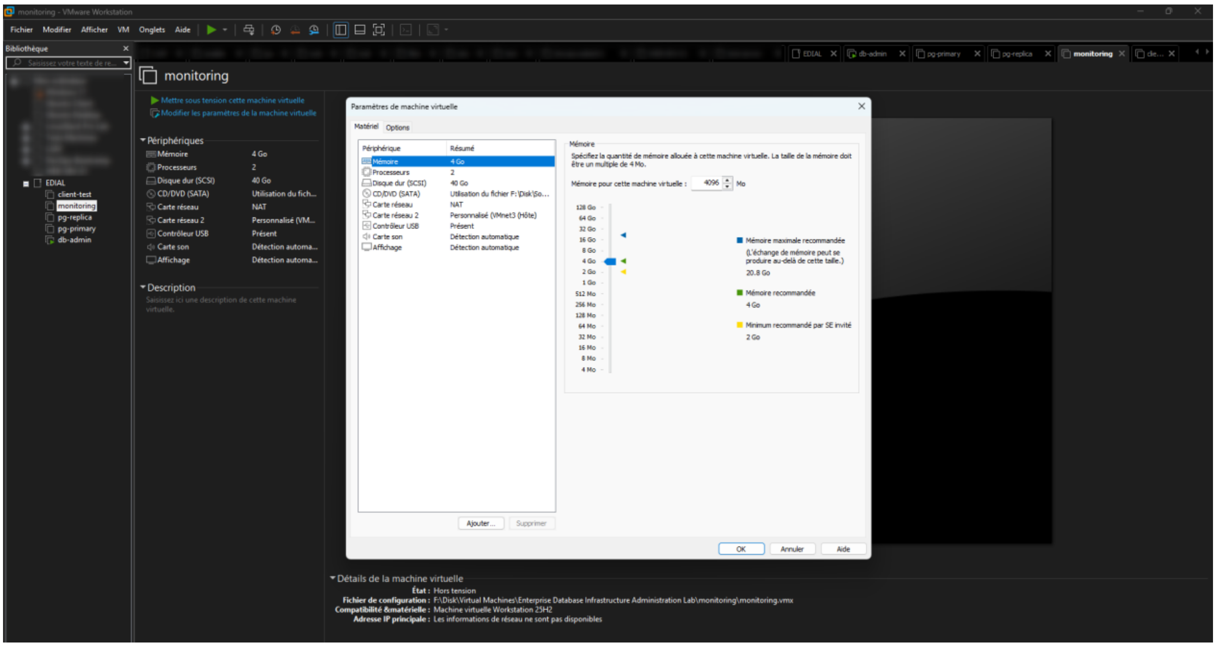
2. pg-primary



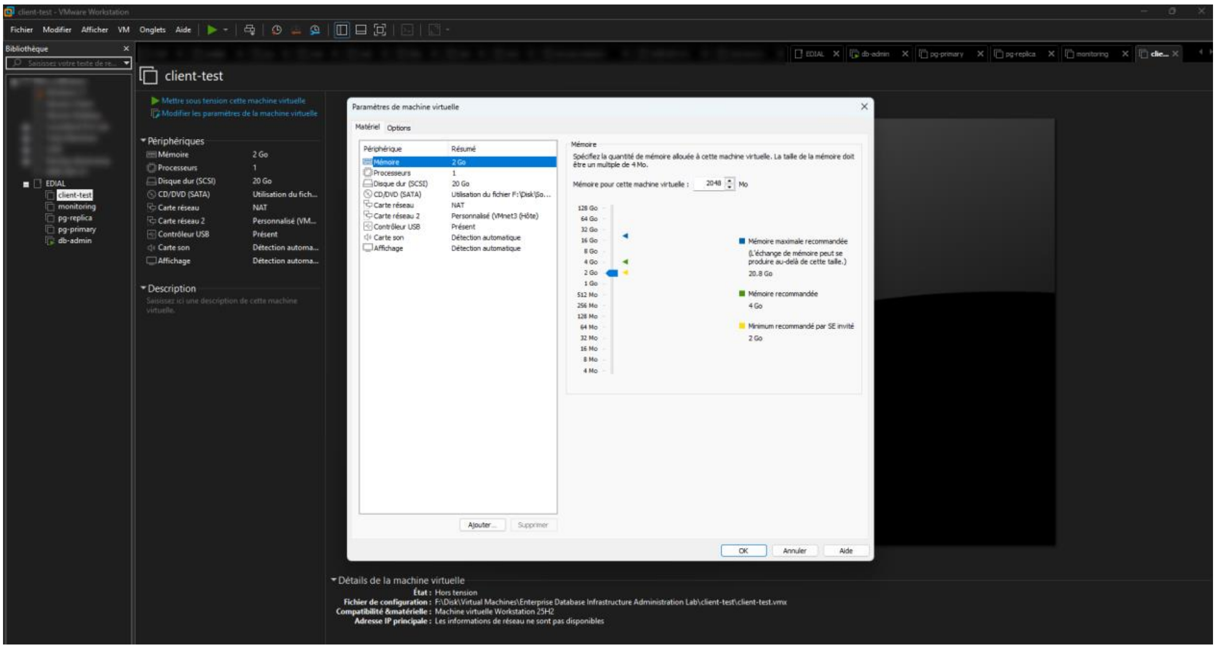
3. pg-replica



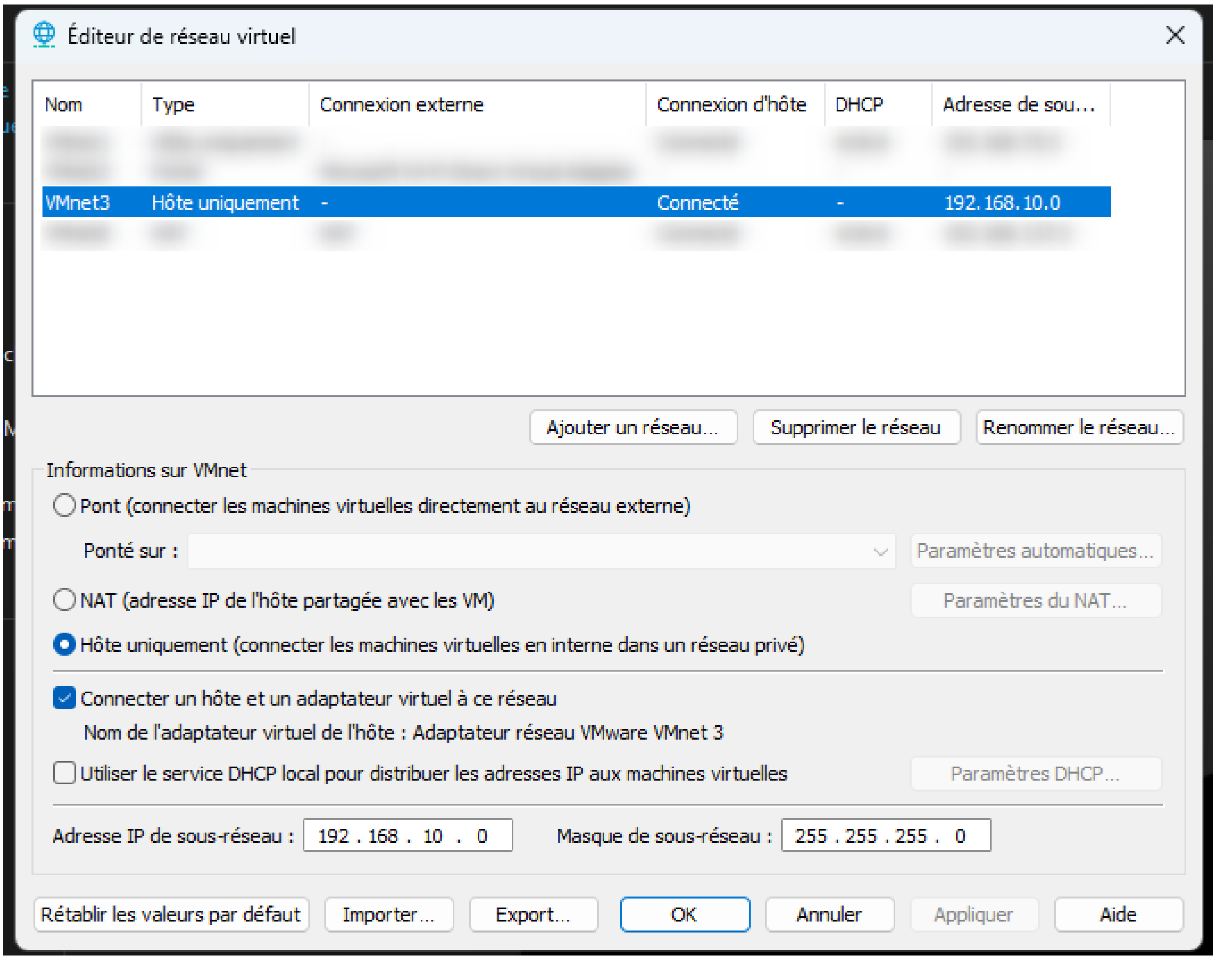
4. monitoring



5. client-test



III. VMware Network Configuration

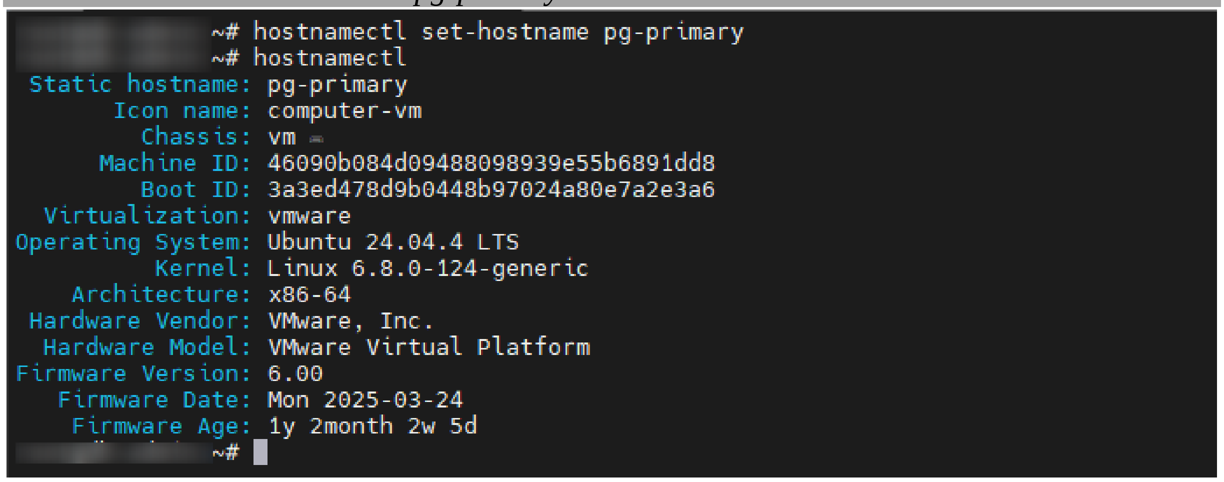


IV. Hostname Configuration

Configure the appropriate hostname on each machine.

Example for pg-primary :

sudo hostnamectl set-hostname pg-primary



V. Static IP Configuration

Ubuntu Server 24.04 uses Netplan.

1. Identify Network Interface

ip a

```
root@db-admin:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:0c:29:e0:c0:b5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.137.153/24 metric 100 brd 192.168.137.255 scope global dynamic ens33
       valid_lft 1007sec preferred_lft 1007sec
    inet6 fe80::20c:29ff:fee0:c0b5/64 scope link
       valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:e0:c0:bf brd ff:ff:ff:ff:ff:ff
    altname enp2s2
root@db-admin:~#
```

2. Configure Netplan

sudo nano /etc/netplan/00-installer-config.yaml

```
2. db-admin
  GNU nano 7.2                                   50-cloud-init.yaml *
network:
  version: 2
  ethernets:
    ens33:
      addresses:
        - 192.168.10.10/24
      routes:
        - to: default
          via: 192.168.10.1
      nameservers:
        addresses:
          - 8.8.8.8
```

3. Apply

sudo netplan apply

VI. Hosts File Configuration

To enable hostname resolution without DNS :*sudo nano /etc/hosts*

```
  GNU nano 7.2                                   /etc/hosts
127.0.0.1 localhost
127.0.1.1 pg-primary


192.168.10.10 db-admin
192.168.10.20 pg-primary
192.168.10.21 pg-replica
192.168.10.30 monitoring
192.168.10.40 client-test


# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

VII. Connectivity Validation

From db-admin :

ping -c 4 pg-primary

ping -c 4 pg-replica

ping -c 4 monitoring

ping -c 4 client-test

```
root@db-admin:~# ping -c 4 pg-primary && ping -c 4 pg-replica && ping -c 4 monitoring && ping -c 4 client-test
PING pg-primary (192.168.10.20) 56(84) bytes of data.
64 bytes from pg-primary (192.168.10.20): icmp_seq=1 ttl=64 time=0.978 ms
64 bytes from pg-primary (192.168.10.20): icmp_seq=2 ttl=64 time=0.676 ms
64 bytes from pg-primary (192.168.10.20): icmp_seq=3 ttl=64 time=0.941 ms
64 bytes from pg-primary (192.168.10.20): icmp_seq=4 ttl=64 time=0.728 ms

--- pg-primary ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3048ms
rtt min/avg/max/mdev = 0.676/0.830/0.978/0.130 ms
PING pg-replica (192.168.10.21) 56(84) bytes of data.
64 bytes from pg-replica (192.168.10.21): icmp_seq=1 ttl=64 time=3.59 ms
64 bytes from pg-replica (192.168.10.21): icmp_seq=2 ttl=64 time=0.788 ms
64 bytes from pg-replica (192.168.10.21): icmp_seq=3 ttl=64 time=0.876 ms
64 bytes from pg-replica (192.168.10.21): icmp_seq=4 ttl=64 time=0.826 ms

--- pg-replica ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.788/1.519/3.586/1.193 ms
PING monitoring (192.168.10.30) 56(84) bytes of data.
64 bytes from monitoring (192.168.10.30): icmp_seq=1 ttl=64 time=1.78 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=2 ttl=64 time=0.700 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=3 ttl=64 time=0.786 ms
64 bytes from monitoring (192.168.10.30): icmp_seq=4 ttl=64 time=0.827 ms

--- monitoring ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3088ms
rtt min/avg/max/mdev = 0.700/1.022/1.775/0.437 ms
PING client-test (192.168.10.40) 56(84) bytes of data.
64 bytes from client-test (192.168.10.40): icmp_seq=1 ttl=64 time=2.86 ms
64 bytes from client-test (192.168.10.40): icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from client-test (192.168.10.40): icmp_seq=3 ttl=64 time=0.918 ms
64 bytes from client-test (192.168.10.40): icmp_seq=4 ttl=64 time=0.724 ms

--- client-test ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.724/1.460/2.855/0.835 ms
root@db-admin:~#
```

VIII. System Update

Run on all servers.

sudo apt update

sudo apt upgrade -y

IX. Install Essential Administration Tools

On all servers :

sudo apt install -y vim curl wget net-tools tree htop unzip zip git rsync tcpdump nmap

X. Baseline Security Hardening

sudo apt install ufw -y

sudo ufw allow 22/tcp

sudo ufw enable

sudo ufw status